

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

CLASS XI

PHYSICS

MECHANICAL PROPERTIES OF FLUIDS

1. Do big or small rain drops fall faster? Why?
2. When air is blown between two balls suspended close to each other, they are attracted towards each other. Why?
3. The clothes are better cleaned with hot water than with cold water. Why?
4. The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. What is the ratio between the volume of the first and the second bubble?
5. A 20 cm long capillary tube is dipped in water. The water rises up to 8 cm. If the entire arrangement is put in a freely falling elevator, what will the length of the water column in the capillary tube be?
6. Water is flowing with a speed of 2 m/s in a horizontal pipe with cross-sectional area decreasing from $2 \times 10^{-2} \text{ m}^2$ to 0.01 m^2 at pressure $4 \times 10^4 \text{ Pa}$. What will the pressure at the small cross-section be?
7. A hydraulic automobile lift is designed to lift cars with a maximum mass of 3000 kg. The area of cross-section of the piston carrying the load is 425 cm^2 . What maximum pressure would the smaller piston have to bear?
8. Calculate the work done in blowing a soap bubble from a radius of 2 cm to 3 cm. The surface tension of the soap solution is 30 dyne cm^{-1} .
9. A mercury drop of radius 1.0 cm is sprayed into 10^6 droplets of equal size. Calculate the energy expended. Surface tension of mercury = $32 \times 10^{-2} \text{ N/m}$.
10. A circular metal plate of radius 5 cm, rests on a layer of castor oil 2 mm thick, whose coefficient of viscosity is 15.5 poise. Calculate the horizontal force required to move the plate with a speed of 5 cm/s.